TRAFFIQ

IET On-Campus JU

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JUMPER

TRAFFIQ

ABOUT THE EVENT

What if a car could see... think... and decide?
TRAFFIQ challenges participants to build an Artificial Intelligence model capable of navigating a vehicle autonomously using real-time camera input.
No remote control.
No manual override.
Just vision, intelligence, and decision-making.
We provide the car.
You build the brain.
This is where computer vision meets control systems.
This is where machine learning meets motion

OBJECTIVE

Your task is to design and deploy an AI model that processes live camera feed, understands the environment, and controls the vehicle to successfully navigate through the arena.
To develop an AI model that:

- Takes real-time image input from an onboard camera
- Processes visual data using computer vision techniques
- Makes steering and speed decisions autonomously
- Navigates the provided car through a predefined track
- The vehicle will be provided by the organizers.
- Teams are responsible for the AI model and control logic.
- There will be obstacles present in the track.

EVENT FORMAT

Each team will be given:

- A standardized vehicle platform
- A Raspberry Pi 4B 8GB
- A mounted Raspberry Pi Cam V2
- Access to control interface for steering and throttle

Teams must:

- Train or design their own AI model
- Integrate it with the vehicle system
- Deploy it during their run
- Navigate the car through the competition track autonomously

The vehicle must operate without any human intervention once the run begins

TEAM SIZE : 2 to 5 members

ROUNDS

Round 1 – Qualifier

- AI model report
- Basic functionality testing

Round 2 – Final

- Complex track with obstacles
- Dynamic elements
- L turns
- Variable lighting conditions

Higher precision requirements

RULES & REGULATIONS

- The vehicle will be provided by the organizers.
- Teams must not modify the hardware of the vehicle.
- Only software/AI-based control is permitted.
- The system must operate fully autonomously once started.
- Manual override during the run, results in penalty or disqualification.
- All AI computation must run on approved hardware.
- Each team must be prepared to briefly explain their AI model architecture, training approach, and decision-making pipeline to the judges upon request.
- Failure to provide a clear explanation may lead to score reduction.
- The model inference time must remain within a specified threshold to ensure real-time performance.
- Teams may not use pre-recorded track footage from the competition arena for training unless explicitly provided by organizers.
- Internet access during the run will not be allowed unless specified.
- Teams are strictly prohibited from using pre-mapped coordinates, fixed path scripts, or hardcoded navigation specific to the competition arena. The navigation must be based purely on real-time visual input.
- The AI system must include a safe-stop mechanism that halts the vehicle in case of model crash, unexpected behavior, or loss of visual input.
- Teams must use only their own developed models or properly credited open-source frameworks. Plagiarism or unauthorized code usage will result in immediate disqualification.
- Once the run begins, no external communication (WiFi commands, Bluetooth override, SSH control, remote desktop, etc.) is permitted.
- The AI model must be robust to moderate lighting changes. The arena lighting may vary slightly during rounds.
- The decision of the organizers will be final and binding.

TRAFFIQ

SAMPLE CHALLENGES

- Fog
- Narrow corridors
- Static obstacles
- Decision-based routing

POINT SYSTEM

Reward Points

- Successful completion of course – 300 points
- Time score – (600 – Time in seconds)
- Obstacle bonus / checkpoints – As declared
- Points on distance covered
- Path optimization efficiency bonus

Penalty Points

- Collision with obstacle – 25 points
- Boundary violation – 20 points
- Human intervention (Maximum 3) – 50 points
- **Hard coding as per arena - Disqualified**

Total Score = (Sum of Reward Points) – (Sum of Penalty Points)

WHAT WE WILL TEST ON

- Computer Vision
- Real-time image processing
- Deep Learning / Machine Learning models
- Path detection and lane recognition
- Obstacle detection
- Decision-making under dynamic conditions
- Model optimization for latency

MODE OF CONDUCT: **1st round: Online** (Report submission)
2nd round: Offline (Arena-based live competition)

PRIZES

1st Prize - 3000
2nd Prize - 2000
3rd Prize - 1000
Innovation Prize - 500
Finalists will receive Certificates of Merit.
All participants will receive Participation e-Certificates.

WHY PARTICIPATE?

Glory. Recognition. Technical dominance.
TRAFFIQ is where:
Mechanical design meets embedded systems
Control theory meets real-world chaos
Algorithms face unpredictability
If you believe your robot can think, adapt, and conquer –
this is your arena.